

Blackboard Contract Draft

Project Drisya Decision Logic

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Draft Blackboard Contract For Decision Logic

This is a draft to discuss with the team. Do not treat it as final until navigation, WiFi detection, simulation, and decision logic owners agree.

Inputs Read By Decision Logic

Key	Owner	Required fields	Notes
mission/control	Simulator/dashboard/open	mode, emergency_stop, timestamp.ms	mode can be idle, run, pause, return, stop.
robot/pose	Navigation/simulation	x, y, theta_deg, timestamp.ms	Coordinates must follow root README convention.
map/occupancy_grid	Navigation	50 x 50 grid	Suggested encoding: unknown -1, free 0, occupied 1, hazard 2.
navigation/path	Navigation	points, status, timestamp.ms	status can be none, planning, ready, blocked, reached.
navigation/frontier	Navigation	target_x, target_y, score, timestamp.ms	Used for exploration baseline.
sensors/proximity	Navigation/simulation	front_cm, left_cm, right_cm, timestamp.ms	Decision logic only needs processed proximity, not raw sensor rays.
detection/result	WiFi detection	human_x, human_y, confidence, timestamp.ms	Confidence should be from 0.0 to 1.0.

Outputs Written By Decision Logic

Key	Consumer	Required fields	Notes
state/motor_command	Simulator/hardware motor layer	left_speed, right_speed, duration_ms, reason, timestamp_ms	Speed range should be agreed, suggested -100 to 100.
decision/state	Dashboard/logs	mission_state, active_behavior, last_transition, timestamp_ms	Used for visualization and debugging.
decision/target	Navigation	x, y, target_type, priority, timestamp_ms	target_type can be victim, frontier, return_home.
decision/trace	Dashboard/logs	tick_id, selected_action, reason, source_layer, timestamp_ms	Can be latest trace entry or a bounded recent list.
detection/confirm_request	WiFi detection/simulation	target_x, target_y, required_confidence, timestamp_ms	Optional, used when confirmation behavior exists.

Example Input Payloads

```
mission_control = {
  "mode": "run",
  "emergency_stop": False,
  "timestamp_ms": 1780000000000,
}
```

```
robot_pose = {
  "x": 12.0,
  "y": 8.0,
  "theta_deg": 90.0,
  "timestamp_ms": 1780000000000,
}
```

```
detection_result = {
  "human_x": 20.0,
  "human_y": 14.0,
  "confidence": 0.82,
  "timestamp_ms": 1780000000000,
}
```

```
proximity = {
  "front_cm": 42.0,
  "left_cm": 55.0,
  "right_cm": 31.0,
  "timestamp_ms": 1780000000000,
}
```

Example Output Payloads

```
motor_command = {
  "left_speed": 0,
  "right_speed": 0,
  "duration_ms": 100,
  "reason": "emergency_stop_requested",
  "timestamp_ms": 1780000000100,
}

decision_state = {
  "mission_state": "stopped",
  "active_behavior": "EmergencyStopSequence",
  "last_transition": "searching -> stopped",
  "timestamp_ms": 1780000000100,
}

decision_target = {
  "x": 20.0,
  "y": 14.0,
  "target_type": "victim",
  "priority": 100,
  "timestamp_ms": 1780000000100,
}

decision_trace = {
  "tick_id": 145,
  "selected_action": "SetVictimTarget",
  "reason": "detection_confidence_0.82_above_threshold_0.75",
  "source_layer": "BT_MISSION",
  "timestamp_ms": 1780000000100,
}
```

Contract Rules

- Decision logic must not act on stale data.
- Decision logic must not move the robot if mission mode is `idle`, `pause`, or `stop`.
- Emergency stop must override all other behavior.
- Obstacle safety must override victim navigation and exploration.
- RL output must be supervised by Behavior Tree safety.
- Every motor command should include a reason.